

A New Blindfold Record?

John Knott reports on the extraordinary achievement of Swedish Grandmaster, Axel Smith

In the year in which Sir David Attenborough has celebrated his 100th birthday, attention is increasingly focusing on environmental issues that affect all life on earth. But even so, who would have imagined that a Swedish campaign to preserve biodiversity in regions threatened by fossil peat mining would have directly led to Grandmaster Axel Smith, a prolific chess author and keen sportsman, attempting to break the world record for simultaneous blindfold games?

We need to step back a little, to put that into context. The campaign is called ‘Vote NO to peat extraction!’ and aims to achieve a ban on peat mining in wetlands crucial for biodiversity and climate mitigation. The campaign highlights that peatlands store significant carbon, and are vital ecosystems that support a large variety of plant and animal species. Axel Smith, when not playing or studying chess, or writing books, makes time within a busy family life to keep fit by running, typically for up to 20 kilometres a day. Understandably, the quality of the countryside is important to him.

The Idea

Earlier in the year it occurred to Axel that he could help draw attention to the campaign through a chess event, and he hit upon the idea of tackling the world record at blindfold simultaneous chess. This was a bold plan. The target that he was aiming to beat was the record of 48 blindfold games played by GM Timur Gareyev in December 2016 in Las Vegas.

Gareyev, then aged 28, took 19 hours (including some breaks) to achieve a result of +35, =7, -6, gaining a score of 80.2 per cent. In preparation for that event, Gareyev, a previous winner of the ever competitive US National Open and the North American Open, and with a peak FIDE rating of 2682, had undertaken a large number of blindfold sessions in the preceding four years, working up to 27, then 33 and then 35 simultaneous games, all with good results (a detailed account of Gareyev’s blindfold history and record performance is given in an article in the blog section of blindfoldchess.net).

In contrast, Axel Smith, whose FIDE rating of 2516 in August 2016 had slipped to 2445 by April of this year, in the past had played no more than 15 blindfold games simultaneously, and that was 15 years ago when he was in his mid-twenties. That was a record for Sweden, but well below not only Gareyev’s 48, but also the earlier total of 46 played by FM Marc Lang in 2011, and the previous record of 45 games played by Miguel Najdorf in 1945.

Not just for the general public, but also for experienced chess players, these were amazing performances, apparently calling for prodigious feats of memory beyond most people’s comprehension. One might have thought that an attempt at 20 blindfold games, to establish a new record for Sweden, would have been sufficient to draw attention to the environmental campaign. However, Axel Smith has a determined, one might say stubborn, streak, and throughout his life he has been prepared to devote great effort to various enterprises, going well beyond limits of endurance that most people would contemplate.

Driven to a Goal

Axel was born in Ystad, southern Sweden, in 1986. When aged eight he was taught chess by his grandfather, and he started going to the local chess club two years later. He developed a love for chess and competed in Skol-SM (the Swedish Junior Championship) each year, usually finishing around places 11 to 20. From the age of 15 he started spending more time on chess, soon to such an extent that he scarcely socialised. He stopped seeing his friends, stopped playing football, and totally focused his spare time on his mission to improve at chess. In consequence, his school grades dropped drastically, but improvement did not come until he started practising with real opponents.

In 2006, being angry at how he had previously played, he contacted Rolf Martens, whom he describes as a legend of Swedish chess, who had won the Swedish Blitz Championship in 1966 and the Swedish Championship the following year, but had rarely played afterwards.

Axel discussed ideas with Martens and started using some of the latter’s offbeat openings. This led to improvement and winning the Swedish Junior Championship in 2006 as the fifth seed. Over a period of 12 months from January 2007 to January 2008, Axel’s rating jumped from 2205 to 2458, although he believes that at that point it was overvalued. He gained the International Master title in 2008 and aimed to become a grandmaster within a year, but he did not realise how far away he was, and it in fact took eight years to obtain the title. He became Nordic Champion in 2013 (achieving his first GM norm), and Swedish Blitz Champion in 2021.

Axel credits much of his progress during those times to interaction with players such as Nils Grandelius, Daniel Semcesen and Aryan Tari, who all became grandmasters too. Additionally, he later followed and helped develop a technique of his friend Hans

Tikkanen – who had risen rapidly from FM in 2007 to GM in 2010, and who has won the Swedish Championship five times – based on repeatedly solving a large collection of tactical puzzles. This gave rise to their best-selling book, *The Woodpecker Method*.

During one training session, Axel studied puzzles for a continuous period of 22 hours in the basement of his home. That experience, and the fitness and stamina involved in running marathons (his best time being just under 2 hours 29 minutes, achieved in 2021) gave him some hope of being able to complete a large number of blindfold games.

Preparation for the Event

In preparing for the challenge, Axel realised that a major problem would be not just remembering the moves, but separating the games in his mind, particularly in the early stages before they had acquired individual characters. This was a factor that had been stressed by all the major blindfold players in the past.

To help with this he considered taking the black pieces in some games, as several other players had done, but he had previously found that when he had done that, he tended to visualise the board sometimes from Black’s side and sometimes from White’s side. This had caused confusion. So he planned to take White in all the games, varying his openings as much as possible. This is a procedure initiated by Pillsbury, a leading blindfold expert at the start of the 20th century, and adopted to at least some degree by all subsequent record holders. Pillsbury gradually increased the record from the year 1900 onwards, and reached 22 games at Moscow in 1902.

Axel’s plan involved 27 different opening strategies for White divided into 1 e4 (8), 1 d4 (6), 1 c4 (2), 1 ♟f3 (1), 1 ♟c3 (2), 1 b3 (1), 1 a3 (1), 1 h4 (1), 1 g4 (1), 1 g3 (1), 1 f4 (1), 1 e3 (1), and 1 b4 (1). For the 1 d4 games he aimed to use lines such as the Trompowsky, the London System and the Barry Attack. He had a comparable plan for the 1 e4 openings. In two of the first 27 games he would play as if with Black, using the opening moves 1 a3 and 1 h4. Then for games 28 to 49, he would start over again.

As those games would have new board numbers, they would call for the use of different images, because using mental imagery to link moves to a board number was a key factor in his preparation. The purpose of the images would be to associate a game and its initial few moves with that particular board, as it would be no use recalling a position

accurately unless it was linked to the correct board number.

Axel intended to create a mental picture for the start of each game, and modify it as each of the first few moves were played. For example, for Game 16, opening with 1 b3, he would create an image of Bent Larsen at the Battle of Verdun. If the opponent replied with 1...♟f6, an elephant would be added to the scene. Instead, the reply 1...♟c6 would cause a donkey to appear. Similarly, modifications would be made for all other responses, and for the subsequent few moves of each side.

By that stage, given the variety of openings, the games should have reached distinct positions, and Axel would be able to continue with a normal memory of each game. He would still be able to associate a game with the correct board number by recalling its initial image, which would trigger a recollection of the way the game developed.

He rehearsed his memory system for several months and, as a severe test of keeping the opening stages of games distinct from each other, in one session he played 49 games against himself without seeing the boards or pieces, continuing each game to eight moves. He thought he had done as much as he could for game separation for the target he had set himself. But that was just a basic step. To play blindfold beyond the opening stage, clearly further processes are required.

How does a blindfold player ‘see’ a position? Memory and Visualisation factors

A mass of evidence over the years indicates that chess masters as a class do not possess exceptional memories outside of chess. So just how is a player able to compete in even one game when blindfold, let alone forty-nine?

In reports of earlier blindfold events, particularly in newspaper accounts, one can find sensational claims that a blindfold player has to remember the contents of all the 64 squares in each game at every move; meaning that for, say, 40 games lasting an average of 30 moves each, he has to remember up to 153,600 pieces of information. At other than a highly superficial level, that is nonsense. It is rather like saying that in remembering a poem we have to consciously recall every letter in every word. In reality, the letters are grouped into words that we recognise, and the words are grouped into phrases and then themes, the text of which becomes self-reinforcing because we perceive its meaning. This, broadly, is a large part of the process of recalling a position in a game of chess.

The stronger the player, and the better a game is played, the easier it is to remember the moves, because each move is logically linked to the next, with some connections (e.g. a capture and a recapture) being more memorable than others. So how much does a blindfold player really need to visualise and remember?

One of the common misconceptions is that blindfold players need to create a mental picture of every square of the board, and



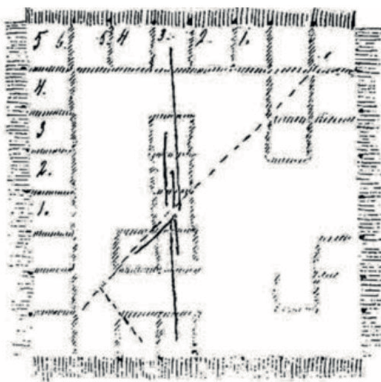
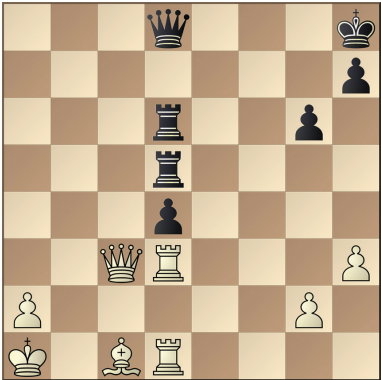
Axel (red singlet) passing the Victory Column in the BMW Berlin Marathon 2021, where he achieved his best time of a very impressive 2:28:47, just 23 minutes behind winner Guye Adola.

all the pieces on those squares, in a form resembling a photograph. That is far from what skilled players actually do, as the effort required to create, maintain, and manipulate even a single complete photographic image would be enormous. Also, it is not necessary.

An experienced blindfold player’s representation of a position is highly abstract, with the degree of abstraction broadly matching his playing strength: the stronger the player, the more abstract will his image typically be. And at any one time, save in a thinly populated endgame, both a player at the board and a blindfold player will typically be envisaging the movement of relatively few of the remaining pieces, and will be concentrating on only the most relevant features. For both, large parts of most current positions will remain unchanged during any but the most intensive analysis.

To give a simplified example – during the first few opening moves of almost any game, neither a sighted nor a blindfold player needs to think about either his own or his opponent’s queen’s rook, or indeed many other static features. During a game, a blindfold player typically creates only as much of an image of the board and pieces as he needs for his immediate purposes, knowing that he can, if necessary, recreate the entire position. It is the spatial disposition of key forces that is important, not the whole board or the appearance of the pieces. The analogy of a spotlight moving among actors on a stage is quite striking: the focus is on only a small area at a time, the rest of the stage being in relative shadow.

Stanislaus Sittenfeld (1865–1902), a Polish-French chess master who played at the Café de la Régence in Paris during the *Belle Époque*, tried to show in a drawing, pictured right, how he might view the position in a blindfold game. Neither the pieces nor the squares form a concrete image. What is held in the mind are the relative locations of the pieces, with their potential relevant activity



As can be seen, the drawing resembles a sketch, or a lightly-drawn cartoon, where only the essential features are present. I leave the reader to work out the sequence of moves that Sittenfeld envisaged, and the result. We can assume that Black’s last move would have beene5xd4.

Over the years there have been a number of investigations into the mental processes involved in playing chess, among which were those carried out by the Dutch psychologist and chess master Adriaan de Groot from the late 1930s onwards. His studies revealed that masters performed much better than experts or weaker players in the task of reconstructing real game positions they had viewed briefly, with the strongest players often reproducing complicated positions without a single error.

De Groot concluded that high-level chess expertise was mostly dependent on the ability of a player to quickly recognise important patterns, and not so much on calculating numerous variations. From the early 1970s onwards, further investigations into perception and memory in chess, by academic psychologists William Chase and Herbert Simon of Carnegie Mellon University, Pittsburg, tended to support de Groot's conclusions. Chase and Simon concentrated on the relevance of groups of pieces (for example, a pawn chain or a castled king's position), which formed easily recognisable 'chunks' of about two to five pieces, whereby one element of a position would actually contain several logically-connected pieces, reducing the demands on memory. They thought that fast pattern recognition led directly to the choice of moves, and that analysing ahead was a minor factor.

Later experiments by other psychologists highlighted shortcomings in that approach, and suggested that to overcome the limitations of short-term memory (typically, around seven distinct items) the chunks created by experienced players must be larger than posited by Chase and Simon, and information must be transmitted faster into long-term memory. But even that modified view failed to account for differences arising from different time controls in a game.

Records of eye-movements showed that, unlike with weaker players, strong players studying positions focused a great deal on *empty* squares, particularly when such squares were affected by many pieces. It was clear that the strong players were rapidly characterising the dynamic qualities of a position as a whole, and were not just remembering static positions, but were rapidly testing them with possible future developments. Such thoughts lead us to wonder what else is needed for playing blindfold.

Beyond Memory and Visualisation

Regardless of their importance, the visualisation and memory components of the skills involved in playing blindfold chess are just the basic requirements. Not only does a player need to grasp and recall positions, he also needs to fight his opponent. As Alekhine – probably the greatest blindfold player of all time* – pointed out:

"If the public at large supposes that the most essential difficulty consists of remembering at all times the relative positions, then it forgets that the blindfold player has to overcome a second, much greater difficulty, namely to fight blindfold, to find in every position, blindfold, the relatively best move!"

How a blindfold player fights will depend on a number of factors including what his motivation is. Some of Alekhine's blindfold games were highly complicated and "wild", but he said that he generally played more cautiously when blindfold, and he would not recognise many of his blindfold games as being his, if he had not known that they were.

Contrary to that approach, George Koltanowski, who established new records for simultaneous blindfold games in 1931 (Antwerp, 30 games) and 1937 (Edinburgh, 34 games), often claimed he could play better and with more energy and originality when he was blindfolded than when he was not. His blindfold games, which include many examples of the ultra-sharp Max Lange Attack, support that claim. How would Axel Smith fare?

When he contacted me during the week before the event Axel thought that his system, outlined above, should work well, but that he was doubtful whether his memory would be up to the task. Compared to previous blindfold simul, he thought that his opponents would be weaker, and there could be substitutions after a few hours when the games (only some of which were due to start with the opponents physically present) would all continue online.

Axel expected to score significantly worse than Gareyev, who he reckoned was better at blindfold chess and who also played faster (Axel had played four games online with Timur Gareyev in 2016, both players being blindfold – the score was 2-2, but he found

that Gareyev was playing faster and generally got better positions). However, he was not so much concerned with what the result would be, as with the publicity that he hoped to generate for the environmental campaign.

Thursday Afternoon

The event began at 15.15 on Thursday, 7th May and finished at 20.15 on Friday, 8th May, giving an overall duration of 29 hours. There was a break of about half an hour from 18.25 on Thursday, and a few subsequent short breaks for eating sandwiches or using the toilet. So probably the actual playing time was about 28 hours. The games started at Vänskapens Hus (the House of Friendships) in Lund, southern Sweden, where a chess club and several other associations meet. Fifteen players were physically present, the remaining games being played over the internet. After the first break, all the games continued online, by which time Axel had walked the two kilometres to his studio for the resumption of the event, mentally repeating the moves in all the games once as he did so.

The first result occurred at board 19 after about four hours' playing time, when the sighted player made a mouse slip (not the only person in the event to do so), and resigned before losing a piece. Next to finish was board 29 where Axel's son Henry (aged 8) was playing. Henry had wanted to play something unusual. His mother, Ellinor, had shown him the basics of the Dutch Defence, but cautioned him not to use it. Nevertheless, he did. Henry succumbed to mate after close to five hours, and in doing so learnt about the hazards of exposed kings and weak diagonals.



The event has just started at Vänskapens Hus. Axel's son Henry is just visible in the far room.

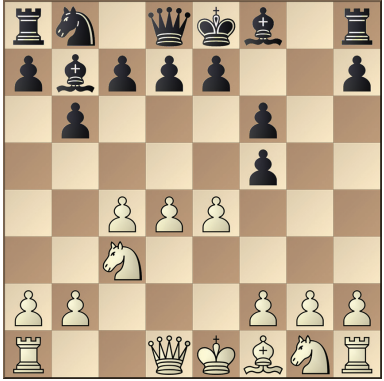
* A. Alekhine, 'The Blindfold Game', Chess Life and Review, vol. 26, pp. 522-523. (A. Buschke, trans. from 'Blindpartien', originally published in Shakhmaty v SSSR, (1931), vol. 14 (July 30), p. 235). Alekhine achieved successive records of 26 games (New York, 1924), 28 (Paris, 1925), and 32 (Chicago, 1933). The opposition at the New York event was the strongest large number successfully faced by any blindfold player.

A.Smith-H.Smith Lund (blindfold simul) 2026 Dutch Defence

1 c4 f5 2 d4 b6 3 ♖c3 ♙b7 4 ♙g5 ♜f6 5 ♙xf6 gxf6?

Recapturing with the e-pawn was preferable, as this leaves Black vulnerable on the h5-e8 diagonal.

6 e4



6...fxe4? 7 ♖h5# 1-0

This was followed by game 33, where after eight hours' playing time the sighted player, rated 1757, mishandled a sharp opening and resigned.

A.Smith-F.Stuhr Lund (blindfold simul) 2026 Scotch Gambit

1 e4 e5 2 ♜f3 ♜c6 3 d4 exd4 4 ♙c4 h6?! 5 0-0 ♙c5 6 c3 dxc3 7 ♙xf7+!

The bishop on c5 is undefended.

7...♙f8

Or 7...♙xf7 8 ♖d5+ ♙f8 9 ♖xc5+ d6 10 ♖c4 with a clear advantage for White.

8 ♜xc3 d6 9 ♙c4 1-0



Black resigned, as although material is equal, his king is badly exposed and he will have difficulty completing development.

All the other games lasted until well after midnight. Next, at board 24, both black horses were put down; at board 30 Axel's daughter Selma (aged 12) unsuccessfully tried the Slav Defence; and at 2.30 on Friday

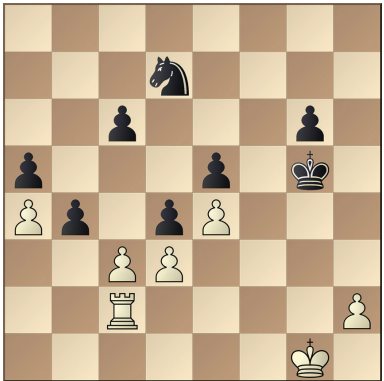


Axel deep in thought after nearly 2 hours' play.

morning, when the position on board 1 was level, GM Ulf Andersson proposed a draw, which was accepted. At that point the results totalled 5½-½ in favour of the player who could not see the boards, giving him hope that the event could be finished successfully.

There was a neat conclusion in game 8, the next to last to finish, where Uno Remitz (1882 Elo) was the exchange down. After 41 moves the following position was reached.

A.Smith-U.Remitz Lund (blindfold simul) 2026



42 cxb4 axb4 43 a5 b3 44 ♙b2 ♜c5 45 ♙xb3!

The point being that if Black captures the rook, his knight will not be able to stop the a-pawn from promoting.

45...♙f4 46 ♙f2 g5 47 ♙e2 ♙g4 48 ♙b6 ♙h3 49 ♙xc6 1-0

A well-fought game on board 14 provided several tactical moments.

A.Smith-A.Bjäre Lund (blindfold simul) 2026 Grob Opening

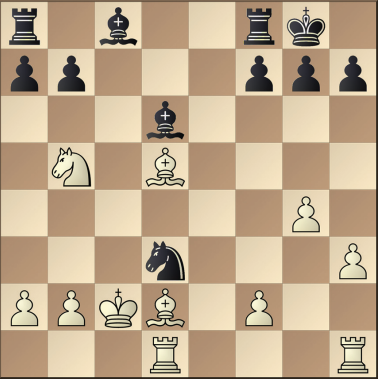
1 g4?! e5 2 ♙g2 d5 3 c4 c6 4 cxd5 cxd5 5 ♙b3 e4

Preferable was 5...♜e7.

6 ♜c3 ♜e7 7 d3 exd3 8 ♜xd5 ♜xd5 9 ♙xd5 ♙d7

Made by the move teller. Sharper was 9...♙b4+ 10 ♙f1! 0-0 with approximate equality.

10 ♙d2 dxe2 11 ♜xe2 ♙d6 12 0-0-0 0-0 13 h3 ♙a6 14 ♜c3 ♜c5 15 ♙c2 ♙c7 16 ♜b5 ♜d3+ 17 ♙b1 ♙xc2+ 18 ♙xc2



18...♜xf2?

Tempting, but correct was 18...♜b4+ 19 ♙xb4 ♙xb4 20 a3 ♙c5 21 b4 ♙e7 22 ♙he1 ♙e6, when Black has freed his queen's rook and has tactical answers to both 23 ♙xe6 and 23 ♙xb7.

19 ♜xd6 ♜xh1 20 ♙xh1

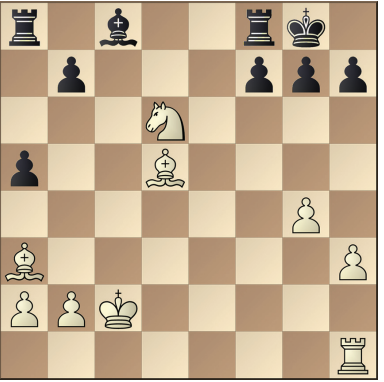
White's position looks a little precarious with the open c- and d-files and an exposed king, but it is all under control.

20...♙d8

Unwisely weakening f7. Best was 20...♙e6.

21 ♙b4 a5 22 ♙a3 ♙f8

Again, 22...♙e6 was relatively best.



23 ♜xf7!

Now, all White's pieces will be in play.

23...xf7

Instead, 23...e8 24 d6+ e6 25 xb7 does not save Black.

24 e1

Also good was 24 f1.

24...d7 25 xf7+ xf7 26 e7+ 1-0

To show how the imagery helped, Axel provided some comments on the initial moves of board 40, where he played the Polish (or Orangutan) Opening against Örjan Hansson.

"The starting picture was a concentration camp (1940) with an orangutan locked inside."

1 b4 e6

"I was playing a Spanish teacher, but apparently he knew French as well – more a story than an image."

2 b2

"Hoping for 2...xb4? 3 xg7, so it's already concrete."

2...f6

"An elephant saying hello to the orangutan."

3 b5

"Logical, moving the threatened pawn. From here I don't think I used any more images."

In games without logical moves, Axel used images for longer. And in hindsight, he believes that he should have used them even more in those games. As another example, he outlined his thoughts during the first few moves of Anderssen's Opening at board 42, where his opponent was August Edström.

"The Norwegian Nazi sympathiser Vidkun Quisling was executed in 1942, and I had prepared (and played) **1 a3**, a move I played once in a tournament in Gothenburg. My mental image was Quisling in that tournament hall." [J.K. – *Actually, Quisling was executed in 1945, but that made no difference as Axel consistently associated the name with board 42.*]

1...b6

"For a fianchetto, my image was a dagger, so I saw Quisling having one to defend himself."

2 d4

"I did think for maybe a minute to decide which move I felt was most logical, and a2-a3 fits in with well many d4 openings. Having taken an informed decision, I was pretty sure to remember it."

2...c5 3 d5



"A Benoni set-up, with my image being bones ('ben' in Swedish), so I had Quisling rotting to bones. Sorry for the disturbing image!"

3...d6



Axel and wife Ellinor at the Malmö Open 2024, an eight-round rapid tournament with 205 players, which Axel won on tie-break with 7 points. Ellinor came 21st with 5½ points.

"I don't remember if I had any image here, however ...d6 was in general a mine."

4 c4

"Again thinking for a minute, avoiding 4 e4 f6 when I would like to play c2-c4 before defending the pawn with d3. With that thought, it was an informed decision."

4...a6 5 e4 b5

"The game has entered a concrete phase, and since move one it has been different from the others, so no more images were needed."

Friday Evening

The last game finished at 20.15 on Friday evening, exactly 29 hours after the start, of which about 28 hours was the actual playing time. Axel Smith was not very tired, and felt as if only a few hours had elapsed, comparable to the duration of a normal tournament game. On the following day he took part in, and won, a 5km race in which 2,147 runners were competing.

Not everyone fared as well. Axel's wife, Ellinor Frisk (a WIM with a peak Elo rating of 2314) had taken one of the boards, but had also acted as the teller for most of the time, relaying moves back and forth. She was helped by IM Isak Storme for three hours, and by her daughter Selma Smith for one hour, each of whom also took a board. Even with that help the task was very tiring. In fact, Ellinor kept falling asleep during the night and had to be woken up when moves were made. In view of the length of the event, many of the sighted players, playing online remotely, also went to sleep and failed to make some moves. When that happened, Ellinor or Isak Storme made a move for them.

There were two games in particular where Axel lost track of part of the position. In one, he agreed a draw, although in reality he had

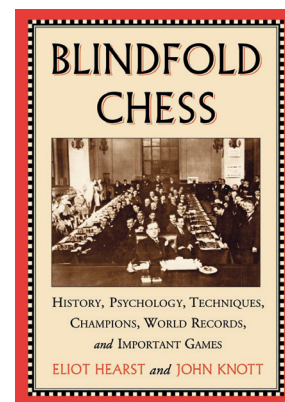
a clearly winning position. And in another he was not sure of the location of his opponent's rook and had to make a couple of waiting moves until the rook moved. Also, there were a few occasions when Axel tried to make a move only to be told by Ellinor that it would be illegal. But overall, particularly considering the number of games and the playing time, the event went as well as could be hoped for.

The final result of the 49 games was 35 wins, 12 draws, and two losses, giving a score of 83.67 per cent. For those interested in statistics, previously Timur Gareyev had taken 19 hours (including interruptions) to play 48 games, scoring 80.2 per cent; and Marc Lange had taken 21 hours, 9 minutes (including interruptions) for 46 games, scoring 75 per cent.

Axel made a total of 1,154 moves during the 28 hours' playing time, giving an average of just over 87 seconds for each move. The average length of the games was 23.55 moves in a range from 7, as shown above, to 49. Part of the length of the event was due to some players failing to resign at the proper time and continuing in clearly lost positions. The event was organised by Restore Wetlands (Återställ Våtmarker) and Disobedient Referendum (Olydig Folkomröstning), and software was provided by ace-chess.com. All the games are available at ace-chess.com/blindsimul and on chess.com.

Purists will no doubt point to the change of playing venue, the young age of at least two of the players, the fact that there were some substitute players, and perhaps other factors, as preventing Axel's achievement from being classed as a new world record. In doing so they will be overlooking, or be in ignorance of, substitutions of players, some weak opponents, and some attempted illegal moves, in several previous events that have long been accepted as creating new records. Regardless of how it comes to be appraised in the future, there is no doubt that this was an amazing performance by Axel Smith.

And how did the vote in the campaign go? That was even better: 97 per cent of the votes were against fossil peat mining. Another success.



Ed. – John and the late Eliot Hearst co-authored the prize-winning and definitive Blindfold Chess (McFarland, 2009), which remains in stock at Chess & Bridge, available for £49.95 or £44.45 for Subscribers.